

## BERKELEY COUNTY AP, SC, USA

WMO: 720598

Lat: 33.183N

Lon: 80.033W

Elev: 22

StdP: 101.06

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00191

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD<br>to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|--------------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                          |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                     | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                      | (o)  | (p)   |
| 1             | -3.0       | -2.0 | -13.6                         | 1.2 | -0.3 | -10.9 | 1.5 | 2.5  | 8.3                   | 17.4 | 7.3 | 14.7 | 1.2                      | 30   | 0.348 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 10.1                   | 35.1            | 25.2 | 33.8 | 24.8 | 32.6 | 24.5 | 26.9                | 31.9 | 26.3 | 30.9 | 25.8 | 30.2 | 2.5                  | 240  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.5                            | 20.7 | 28.4 | 25.1 | 20.3 | 28.2 | 24.8 | 19.9 | 27.9 | 83.8          | 31.4 | 81.9 | 31.0 | 79.8 | 30.2 | 29.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |  |  |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|--|--|
| 1%                | 2.5% | 5%  | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |  |  |
| (a)               | (b)  | (c) | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |  |
| 6.9               | 5.6  | 4.9 | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |  |
|                   |      |     | -6.8                       | 36.6 | 2.6                | 1.8 | -8.7                                               | 37.9 | -10.2      | 38.9 | -11.7      | 39.9 | -13.6      | 41.2 |  |  |
|                   |      |     | -7.9                       | 27.5 | 2.5                | 1.0 | -9.7                                               | 28.2 | -11.1      | 28.8 | -12.5      | 29.4 | -14.3      | 30.1 |  |  |
|                   |      |     | DB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |
|                   |      |     | WB                         |      |                    |     |                                                    |      |            |      |            |      |            |      |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 18.8   | 8.3   | 11.0  | 14.4  | 18.4  | 22.9  | 26.3  | 27.6  | 27.1  | 24.8  | 19.4  | 13.3  | 11.2  |
|                                                                        | DBStd       | 7.73   | 5.67  | 5.14  | 4.89  | 3.93  | 3.00  | 2.44  | 1.77  | 1.58  | 2.20  | 3.84  | 4.50  | 5.01  |
|                                                                        | HDD10.0     | 226    | 101   | 46    | 17    | 1     | 0     | 0     | 0     | 0     | 0     | 0     | 16    | 46    |
|                                                                        | HDD18.3     | 1140   | 313   | 210   | 139   | 48    | 4     | 0     | 0     | 0     | 0     | 34    | 162   | 228   |
|                                                                        | CDD10.0     | 3421   | 49    | 73    | 153   | 252   | 401   | 489   | 544   | 529   | 443   | 292   | 114   | 82    |
|                                                                        | CDD18.3     | 1293   | 2     | 4     | 17    | 49    | 147   | 239   | 286   | 271   | 193   | 68    | 11    | 6     |
|                                                                        | CDH23.3     | 11513  | 13    | 38    | 156   | 433   | 1198  | 2311  | 2870  | 2473  | 1476  | 472   | 54    | 19    |
| (13)                                                                   | CDH26.7     | 4561   | 1     | 3     | 32    | 103   | 417   | 1040  | 1300  | 1023  | 532   | 105   | 3     | 1     |
| (14)                                                                   | Wind        | WSAvg  | 1.8   | 1.8   | 2.1   | 2.2   | 2.3   | 1.9   | 1.6   | 1.7   | 1.4   | 1.5   | 1.7   | 1.7   |
| (15) Precipitation                                                     | PrecAvg     | 1193   | 85    | 91    | 93    | 77    | 86    | 131   | 135   | 149   | 104   | 89    | 67    | 86    |
|                                                                        | PrecMax     | 1472   | 184   | 209   | 194   | 159   | 173   | 251   | 277   | 276   | 253   | 337   | 186   | 186   |
|                                                                        | PrecMin     | 874    | 34    | 36    | 12    | 23    | 29    | 64    | 66    | 82    | 7     | 0     | 4     | 21    |
|                                                                        | PrecStd     | 176    | 45    | 42    | 40    | 34    | 37    | 47    | 52    | 48    | 51    | 75    | 45    | 40    |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 25.1  | 27.2  | 29.8  | 31.2  | 35.0  | 37.3  | 37.3  | 35.9  | 33.0  | 31.2  | 27.3  | 25.9  |
|                                                                        |             | MCWB   | 18.7  | 18.9  | 18.2  | 20.7  | 23.8  | 24.9  | 25.6  | 26.1  | 24.3  | 24.6  | 21.2  | 20.6  |
|                                                                        | 2%          | DB     | 22.3  | 24.1  | 27.4  | 29.1  | 32.4  | 35.0  | 35.1  | 33.8  | 32.3  | 29.0  | 25.0  | 23.0  |
|                                                                        |             | MCWB   | 17.8  | 17.5  | 18.1  | 19.8  | 22.6  | 24.8  | 25.4  | 25.3  | 24.1  | 23.2  | 19.4  | 19.0  |
|                                                                        | 5%          | DB     | 19.7  | 22.3  | 24.9  | 27.6  | 30.9  | 33.0  | 33.8  | 32.6  | 31.2  | 27.6  | 22.8  | 21.4  |
|                                                                        |             | MCWB   | 15.9  | 16.0  | 16.9  | 19.0  | 21.9  | 24.1  | 25.2  | 25.0  | 24.0  | 21.8  | 18.7  | 18.4  |
|                                                                        | 10%         | DB     | 17.5  | 19.8  | 22.6  | 26.1  | 28.9  | 32.2  | 32.5  | 32.0  | 29.9  | 26.2  | 21.2  | 19.8  |
|                                                                        |             | MCWB   | 14.5  | 15.2  | 16.0  | 18.5  | 21.8  | 24.1  | 25.0  | 24.9  | 23.7  | 20.9  | 18.2  | 17.7  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 20.4  | 20.4  | 20.6  | 23.4  | 24.9  | 26.8  | 27.8  | 27.5  | 26.9  | 25.6  | 22.9  | 21.8  |
|                                                                        |             | MCDB   | 22.2  | 24.2  | 25.4  | 27.1  | 30.8  | 33.3  | 33.6  | 33.0  | 30.3  | 28.4  | 25.7  | 24.3  |
|                                                                        | 2%          | WB     | 18.8  | 18.9  | 19.6  | 22.2  | 24.0  | 26.0  | 26.9  | 26.8  | 26.0  | 24.6  | 21.3  | 20.4  |
|                                                                        |             | MCDB   | 21.5  | 22.5  | 24.1  | 26.5  | 29.6  | 31.8  | 32.1  | 31.7  | 29.4  | 27.6  | 22.9  | 22.1  |
|                                                                        | 5%          | WB     | 17.2  | 17.5  | 18.6  | 21.2  | 23.3  | 25.5  | 26.3  | 26.2  | 25.4  | 23.7  | 20.2  | 19.1  |
|                                                                        |             | MCDB   | 18.7  | 20.5  | 22.8  | 25.2  | 28.3  | 30.8  | 31.0  | 30.5  | 28.9  | 26.5  | 21.9  | 20.8  |
|                                                                        | 10%         | WB     | 15.1  | 16.0  | 17.4  | 20.1  | 22.7  | 24.9  | 25.8  | 25.7  | 24.7  | 22.6  | 18.4  | 17.5  |
|                                                                        |             | MCDB   | 17.3  | 18.9  | 21.3  | 23.8  | 27.2  | 30.0  | 30.2  | 29.8  | 28.0  | 25.0  | 21.1  | 18.9  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 10.9  | 11.9  | 12.4  | 12.6  | 11.1  | 10.5  | 10.1  | 9.3   | 9.5   | 11.0  | 11.7  | 10.6  |
|                                                                        |             | MCDBR  | 12.9  | 14.2  | 14.6  | 14.3  | 13.0  | 11.7  | 11.4  | 10.6  | 11.0  | 11.7  | 12.7  | 12.1  |
|                                                                        |             | MCWBR  | 8.6   | 8.2   | 6.7   | 6.3   | 4.9   | 4.1   | 3.9   | 3.9   | 4.2   | 6.1   | 7.8   | 8.4   |
|                                                                        | 5% WB       | MCDBR  | 10.7  | 12.7  | 12.0  | 10.8  | 10.7  | 9.9   | 9.2   | 8.9   | 9.5   | 10.0  | 10.5  | 10.5  |
|                                                                        |             | MCWBR  | 9.0   | 8.8   | 6.6   | 5.7   | 4.5   | 4.3   | 4.1   | 3.9   | 3.9   | 5.7   | 7.5   | 8.1   |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.323 | 0.331 | 0.365 | 0.408 | 0.473 | 0.503 | 0.545 | 0.520 | 0.436 | 0.391 | 0.345 | 0.326 |
|                                                                        | taud        |        | 2.530 | 2.503 | 2.428 | 2.291 | 2.156 | 2.122 | 2.020 | 2.084 | 2.315 | 2.415 | 2.505 | 2.553 |
|                                                                        | Ebn at Noon |        | 896   | 924   | 912   | 882   | 824   | 796   | 760   | 773   | 829   | 844   | 861   | 872   |
|                                                                        | Edh at Noon |        | 84    | 95    | 110   | 132   | 152   | 157   | 173   | 160   | 122   | 102   | 85    | 78    |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 2.83  | 3.48  | 4.60  | 5.86  | 6.31  | 6.45  | 6.19  | 5.58  | 4.77  | 4.08  | 3.14  | 2.52  |
|                                                                        | RadStd      |        | 0.16  | 0.35  | 0.40  | 0.39  | 0.45  | 0.45  | 0.39  | 0.32  | 0.40  | 0.46  | 0.23  | 0.21  |

## Historical Trends

|                         | DBAvg      | Heating    |            | Cooling    |            |            | Degree-Days |            |            |            |
|-------------------------|------------|------------|------------|------------|------------|------------|-------------|------------|------------|------------|
|                         |            | 99% DB     | 99% DP     | 1% DB      | 1% WB      | 1% DP      | HDD10.0     | HDD18.3    | CDD10.0    | CDD18.3    |
| Station Only            | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A  | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A |
| Regional (48 neighbors) | +0.59      | N/A        | N/A        | N/A        | N/A        | N/A        | N/A         | N/A        | +201       | +142       |

Nomenclature: See separate page